

No.	Solution	Remarks
6(a)	Photoelectric effect is the emission of electrons from a metal surface when it is exposed to electromagnetic radiation of sufficiently high frequency.	[1]
6(b)	Incident power, $P = \left(\frac{N}{t} \right) hf$ $\therefore \frac{N}{t} = \frac{P}{hf}$ $= \frac{3.8 \times 10^{-3}}{4.7 \times 1.60 \times 10^{-19}}$ $= 5.05 \times 10^{15} \text{ s}^{-1}$	[1] correct substitution [1] correct answer
6(c)	Photoelectric current, $I = \left(\frac{N'}{t} \right) e$ From Fig. 8.1, max current = $0.80 \times 10^{-8} \text{ A}$ $\left(\frac{N'}{t} \right) = \frac{I}{e}$ $= \frac{0.80 \times 10^{-8}}{1.60 \times 10^{-19}}$ $= 5.0 \times 10^{10} \text{ s}^{-1}$	[1] correct substitution [1] correct answer
6(d)	<u>Rate of incident photons is very much larger than the rate of electrons emitted.</u> <u>Most of the photons rebound from the metal surface, without being absorbed by the electrons.</u>	[1] [1]
6(e)	From Fig. 8.1, stopping potential $V_s = -0.50 \text{ V}$. Maximum KE of electrons, $\text{KE}_{\text{max}} = eV_s$ $= 0.50 \times 1.60 \times 10^{-19}$ $= 8.0 \times 10^{-20} \text{ J}$	[1] correct substitution [1] correct answer
6(f)(i)	Work function of the surface, ϕ , refers to the <u>minimum amount of energy needed to liberate an electron</u> from the metal surface.	[1]
6(f)(ii)	Work function, $\phi = hf - \text{KE}_{\text{max}}$ $= 4.7 - 0.5$ $= 4.2 \text{ eV}$	[1] correct answer

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